MicroBT Whatsminer Diagnostic Checklist

Technician Name: ___________________________ Date: __________________
Miner Model: _____________________________ Serial Number: __________________
Firmware Version: _________________________ Reported Issue: __________________
Customer/Location: ______________________ Ticket #: __________________

1. External Inspection

- ☐ Check for physical damage to case/chassis
- ☐ Inspect power cables for damage or fraying
- ☐ Check the power supply unit for physical damage
- ☐ Inspect the network cable and connection
- ☐ Look for signs of overheating (discoloration, melting)
- ☐ Check for unusual odors (burnt electronics)
- ☐ Inspect fans for dust accumulation or damage
- ☐ Verify all screws and fasteners are present
- ☐ Inspect air intake and exhaust areas for blockage
- ☐ Check for any liquid damage or residue
- ☐ Inspect external connectors for corrosion or damage

2. Initial Power Supply Assessment

- ☐ Verify power button functionality (if applicable)
- ☐ Check power supply output connections
- ☐ Measure power supply output voltages with multimeter:
    - ☐ Main voltage: ________ V (Target: 11.8-12.2V)
    - ☐ Secondary voltage: ________ V (if applicable)
- ☐ Check PSU fan functionality
- ☐ Inspect PSU internal components if suspect (qualified personnel only)
- ☐ Record PSU model/rating: __________________
- ☐ Take photos of any damage or abnormalities

3. Internal Hardware Inspection

- ☐ Remove covers following proper ESD procedures
- ☐ Inspect the control board for:
    - ☐ Burnt or damaged components
    - ☐ Bulging or leaking capacitors
    - ☐ Corrosion or residue
    - ☐ Cold solder joints
    - ☐ Broken traces
    - ☐ Crystal oscillator condition
    - ☐ PIC microcontroller condition
- ☐ Inspect all hash boards for:
    - ☐ Burnt ASIC chips
    - ☐ Damaged components
    - ☐ Corrosion on connectors
    - ☐ Physical damage to board
    - ☐ Burned traces or areas
    - ☐ Proper installation in slots
- ☐ Check all internal connections and cables:
    - ☐ Data cables properly seated
    - ☐ Power connectors secure
    - ☐ Ribbon cables (if present) undamaged
- ☐ Inspect heat sinks and thermal pad/paste condition
- ☐ Check fan connectors and wiring

4. Powered-On Diagnostics (Safe Mode)

- ☐ Connect power (not to the mining pool)
- ☐ Observe startup sequence:
    - ☐ Fan startup and operation
    - ☐ LED indicator patterns
    - ☐ Listen for unusual sounds
    - ☐ Startup time normal/abnormal
- ☐ Record fan behavior:
    - ☐ All fans operating: Yes/No/Partial (list non-functioning fans)
    - ☐ Fans at proper speed: Yes/No
    - ☐ Unusual noise: Yes/No (describe: ____________________)
- ☐ Observe control board LED indicators:
    - ☐ Power LED: On/Off/Flickering
    - ☐ Status LED: On/Off/Flickering

- [ ] Network LED: On/Off/Flickering
- [ ] Hash board status LEDs: Notes: _________________

5. Network Connectivity Check

- [ ] Connect the Ethernet cable
- [ ] Verify the link and activity LEDs are active
- [ ] Scan the network to find miner's IP or check DHCP server
- [ ] Record IP address: _________________
- [ ] Ping miner IP address: Success/Fail
- [ ] Access web interface: Success/Fail
- [ ] Login to the web interface: Success/Fail
- [ ] Check firmware upgrade availability

6. Web Interface Analysis

- [ ] Record temperature readings:
  - [ ] Control board: _____ °C
  - [ ] Hash board 1: _____ °C
  - [ ] Hash board 2: _____ °C
  - [ ] Hash board 3: _____ °C (M30/M31/M50 models)
  - [ ] Hash board 4: _____ °C (M50 models if applicable)
- [ ] Check hash board detection:
  - [ ] Number of boards detected: _____/3 or 4
  - [ ] Chips detected per board: /
  - [ ] Number of operating domains per board: /
- [ ] Check system logs for errors:
  - [ ] Kernel log error messages: _________________
  - [ ] Mining log error messages: _________________
- [ ] Record system uptime: _________________
- [ ] Verify current firmware version: _________________
- [ ] Check power consumption data: _______ W
- [ ] Check chip frequency settings: _______ MHz
- [ ] Verify miner configuration parameters

7. Hash Board Diagnostics

For each hash board (typically 2, 3, or 4 boards depending on model):

Hash Board 1

- ☐ Visual inspection complete
- ☐ Check voltage at test points:
  - ☐ Core voltage: _______ V (Target: Model-specific)
  - ☐ IO voltage: _______ V (Target: Model-specific)
- ☐ Number of working domains detected: /
- ☐ Number of working chips detected: /
- ☐ Hash rate reported: _______ TH/s
- ☐ Operation stable for 10+ minutes: Yes/No
- ☐ Temperature within specs (<85°C typical): Yes/No
- ☐ Board-specific issues: _________________

Hash Board 2

- ☐ Visual inspection complete
- ☐ Check voltage at test points:
  - ☐ Core voltage: _______ V (Target: Model-specific)
  - ☐ IO voltage: _______ V (Target: Model-specific)
- ☐ Number of working domains detected: /
- ☐ Number of working chips detected: /
- ☐ Hash rate reported: _______ TH/s
- ☐ Operation stable for 10+ minutes: Yes/No
- ☐ Temperature within specs (<85°C typical): Yes/No
- ☐ Board-specific issues: _________________

Hash Board 3

- ☐ Visual inspection complete
- ☐ Check voltage at test points:
  - ☐ Core voltage: _______ V (Target: Model-specific)
  - ☐ IO voltage: _______ V (Target: Model-specific)
- ☐ Number of working domains detected: /
- ☐ Number of working chips detected: /
- ☐ Hash rate reported: _______ TH/s
- ☐ Operation stable for 10+ minutes: Yes/No
- ☐ Temperature within specs (<85°C typical): Yes/No
- ☐ Board-specific issues: _________________

Hash Board 4 (if applicable)

- ☐ Visual inspection complete
- ☐ Check voltage at test points:
  - ☐ Core voltage: _______ V (Target: Model-specific)
  - ☐ IO voltage: _______ V (Target: Model-specific)
- ☐ Number of working domains detected: /
- ☐ Number of working chips detected: /
- ☐ Hash rate reported: _______ TH/s
- ☐ Operation stable for 10+ minutes: Yes/No
- ☐ Temperature within specs (<85°C typical): Yes/No
- ☐ Board-specific issues: _________________

8. Special Diagnostic Tests

- ☐ Run manufacturers' diagnostic tools if available
- ☐ Perform resistor tests on suspicious component groups
- ☐ Check PIC controller functionality (if applicable)
- ☐ Verify frequency settings in all domains
- ☐ Test power regulation circuit operation
- ☐ Check communication between control board and hash boards

9. Performance Testing

- ☐ Connect to test mining pool (if authorized)
- ☐ Run for minimum 30 minutes
- ☐ Record stable hash rate: _______ TH/s
- ☐ Record expected hash rate for model: _______ TH/s
- ☐ Performance percentage: _______% (Target: >95%)
- ☐ Record reject rate: _______% (Target: <1%)
- ☐ Record power consumption: _______ W
- ☐ Calculate efficiency: _______ J/TH
- ☐ Thermal stability: Pass/Fail
- ☐ Fan performance under load: Pass/Fail
- ☐ Network stability during test: Pass/Fail

10. Firmware Assessment

- ☐ Current firmware appropriate for model: Yes/No
- ☐ Current firmware version: _________________
- ☐ Latest available firmware version: _________________
- ☐ Firmware update required: Yes/No
- ☐ Recovery mode functioning: Yes/No
- ☐ Configuration settings appropriate: Yes/No
- ☐ Security settings configured properly: Yes/No

11. Model-Specific Checks

For M20/M21 Series

- ☐ Check crystal oscillator functionality
- ☐ Verify temperature sensor operation
- ☐ Check data cable connection to hash boards
- ☐ Verify voltage regulation for each domain

For M30/M31 Series

- ☐ Check ribbon cable connections and integrity
- ☐ Verify power distribution across hash boards
- ☐ Test proper fan speed regulation
- ☐ Check advanced temperature management

For M50 Series

- ☐ Verify cooling system operation (including liquid cooling if applicable)
- ☐ Check the advanced power management system
- ☐ Test high-density board connections
- ☐ Verify firmware compatibility with hardware revision

12. Problem Diagnosis Summary

- ☐ Primary issue identified: ____
- ☐ Secondary issues identified: ____
- ☐ Root cause assessment: ____
- ☐ Components requiring repair/replacement:
    - ☐ ____
    - ☐ ____

13. Repair Plan

- ☐ Recommended actions:
  - ☐ ______________________________
  - ☐ ______________________________
  - ☐ ______________________________
- ☐ Estimated repair time: _________________
- ☐ Parts required: ____
- ☐ Estimated cost: _________________
- ☐ Repair priority: High / Medium / Low

14. Additional Notes

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15. Final Recommendations

- ☐ Proceed with repairs
- ☐ Beyond economic repair - recommend replacement
- ☐ Partial repair recommended
- ☐ Additional diagnosis required
- ☐ Return to manufacturer (RMA)
- ☐ No issues found - return to service
- ☐ Firmware update only
- ☐ Preventative maintenance recommended

Diagnostic Completed By: ____

Signature: ____

Date/Time Completed: _________________

Repair Approved By:____
Date: _________________

